

Sim Timing Field Guide

What one row means, which numbers you can trust, and the five things that will bite you if you chart this data without reading first.

62 columns

1 row per simulation

key `UniqueRowID`

site `Lacey CT_Sim`

floor `2025-10-01`

WHAT ONE ROW IS

One **completed** simulation appointment on the Lacey CT_Sim scanner **that has at least one CT series matched to it**. Every column on the row describes that single appointment — there is no fan-out, so counts and averages need no de-duplication.

DO NOT COUNT ROWS TO ANSWER "HOW MANY SIMS DID WE DO"

The report requires matched CT imaging, so real completed simulations that lack an attributed scan are **absent, not blank**. For recent months the loss is small — the window evidence puts matching at roughly 110/112 and 113/116 — but it is not zero and it is not random. Sim volume comes from `Simulations.sql`. This report answers a narrower question: *for the sims we can see the CT for, how long did they take*.

JOINING OUT

`UniqueRowID` is the same key `Simulations.sql` emits under that name, so the two join **1:1 with no transformation**. Join `Simulations.sql` → left, this → right; expect unmatched `Simulations` rows for Centralia, electron CSU, non-completed sims, and sims with no attributed CT.

The CT simulator's clock ran **75–90 minutes slow for over two years** and was corrected the weekend of 2026-07-25/26. It was the scanner, not the warehouse — linac imaging held a steady –3 min through the same period. The error also *drifted*, losing a further 1.5–2 min per month, which is why no single offset repairs it.

READ CTTIMESTAMPERA ON EVERY ROW BEFORE PLOTTING A CT TIME

'**Reliable**' means the appointment is on or after 2026-07-27. Anything else is prefix, and its CT timestamps have been **algorithmically corrected** using a per-month, per-era offset — **CTClockOffsetMinutes** tells you exactly how many minutes were added to that row.

Corrected values are sound for **population work** — monthly medians, distributions, trends. They are **not** sound per-appointment: residual scatter is ±10–20 minutes. Never use a corrected row to adjudicate an individual case.

What the clock touched and what it never did		
STATUS	COLUMNS	WHY
SAFE ALWAYS	ScheduledDurationMinutes , StartDelayMinutes , ActualDurationMinutes	Built only from appointment stamps. The scanner clock never touched them.
SAFE ALWAYS	CTScanSpanMinutes	Both endpoints come from the scanner's own clock, so the offset cancels exactly.
SAFE ALWAYS	StudyCount , SeriesCount , all note columns	Counts and ARIA-side timestamps, unaffected by the scanner.
CROSS-CLOCK	CheckInToCTCompleteMinutes ,	These subtract a

ScheduledToCTCompleteMinutes ,
CTCompleteToApptEndMinutes ,
CTCompleteToScheduledEndMinutes ,
CTCompleteToDocCompleteMinutes , and
CTCompletedDateTime as an absolute time

scanner
timestamp from an
ARIA one. Gate on
CTTimestampEra .

THE COLUMNS

Grouped by lineage — identity, the appointment's own clock, the scanner, the notes, then the end-to-end measures that combine them. Every interval endpoint is emitted as a timestamp, so any interval not listed is one subtraction away in pandas.

Identity who, what, where

COLUMN	TYPE	NOTES
UniqueRowID	int	Primary key. Joins 1:1 to <code>Simulations.sql</code> .
PatientId	str	MRN. PHI
PatientFullName	str	PHI Drop before sharing any export.
ActivityName	str	e.g. Initial Simulation, Stereotactic. Good natural facet.
ActivityStatus	str	Always one of three completed states — near-constant, low value as a facet.
SimulationResource	str	Constant (<code>CT_Sim</code>) while the report is Lacey-only.
DepartmentName	str	Effectively constant for the same reason.
CompletedByUser	str	Who closed the appointment. Useful for per-therapist views.
AppointmentNote	str	Free text. PHI Not structured — don't parse it.

The appointment clock ARIA-side, trustworthy across all history

COLUMN	TYPE	NOTES
ScheduledStartDateTime	datetime	The booked start. Your primary time axis.

ScheduledEndTime	datetime	The booked end.
CheckedInFlag	str	ARIA's own check-in flag.
CheckInDateTime	datetime?	Arrival. NULL when never checked in — that is real, not missing data.
ActualEndTime	datetime?	When the appointment was closed out.
ScheduledDurationMinutes	int	Booked slot length.
StartDelayMinutes	int?	Scheduled start → check-in. Positive = late.
ActualDurationMinutes	int?	Check-in → actual end.

The scanner gate the cross-clock ones on CTTimestampEra

COLUMN	TYPE	NOTES
CTFirstSeriesDateTime	datetime	First series landed. Clock-corrected when pre-fix.
CTCompletedDateTime	datetime	Headline. Last series landed — closest thing ARIA has to "scan finished".
CTTimestampEra	str	'Reliable' or an UNRELIABLE string. Filter on this.
CTClockOffsetMinutes	int	Minutes added to this row's CT stamps. 0 post-fix.
StudyCount	int	Distinct studies assigned.
SeriesCount	int	Distinct series. Not slices — true slice counts don't exist in this warehouse.
CTScanSpanMinutes	int	First → last series. Clock-immune, valid across all history.
CheckInToCTCompleteMinutes	int?	The patient-facing scan experience. CROSS-CLOCK
ScheduledToCTCompleteMinutes	int?	CROSS-CLOCK
CTCompleteToApptEndMinutes	int?	Teardown tail after the scan. CROSS-CLOCK

CTCompleteToScheduledEndMinutes	int?	Negative = ran past the booked slot. CROSS-CLOCK
CTLastImageId	str	
CTImageType	str	
CTImageStatus	str	
CTSliceRTType	str	
CTDicomUID	str	

The notes Simulation Note and Simulation Set Up Notes only

COLUMN	TYPE	NOTES
SimNote_DateOfService	datetime?	User-editable. Not an event time — can be backdated.
SimNote_DateEntered	datetime?	The trustworthy one. System-set, never moves.
SetupNote_DateOfService	datetime?	Same caveat as above.
SetupNote_DateEntered	datetime?	System-set.
SetupNote_LastModified	datetime?	Set Up Notes only — physicians never touch them, so this is genuinely the author. There is deliberately no sim-note equivalent.
SimNote_WriteSeconds	int?	Authoring time, ~79–122 s typical. Screen outliers — see gotchas.
SetupNote_WriteSeconds	int?	Same.
SetupNote_WriteSecondsToLastModified	int?	Includes a reopen-and-amend. $\geq$ SetupNote_WriteSeconds .
BothNotes_WriteSecondsTotal	int?	Combined authoring across both notes.
HasSimOrSetupNote	0/1	Any in-scope note attached.

LatestSimOrSetupNoteType	str	Whichever of the two types landed last, with its two timestamps.
LatestSimOrSetupNoteDateTime	datetime?	
LatestSimOrSetupNote_DateEntered	datetime?	
MinutesCTCompleteToSimNoteEntered	int?	Documentation lag after the scan. CROSS-CLOCK
MinutesCTCompleteToSetupNoteEntered	int?	CROSS-CLOCK
MinutesCheckInToSimNoteEntered	int?	ARIA-side both ends — safe across all history.
SimNote_LinkMethod	str	Reads Patient+window on essentially every row. Expected, not a defect — no exact note↔appointment key exists.
SetupNote_LinkMethod	str	
SimSetupNotesFound	int	In-scope notes attached to this appointment.
SimSetupNotesViaActivityLink	int	Near-always 0. See link method above.
SimSetupNotesWithSignedDate	int	Expected 0 — this site doesn't sign these notes.

Visit + documentation, end to end
what the report exists to measure

COLUMN	TYPE	NOTES
DocumentationCompleteDateTime	datetime?	Last system stamp any in-scope note carries.
CheckInToDocCompleteMinutes	int?	The headline metric. True end-to-end burden of the visit.
ScheduledStartToDocCompleteMinutes	int?	Same, anchored on the booked start — includes patient lateness.
CTCompleteToDocCompleteMinutes	int?	Documentation tail alone. CROSS-CLOCK

DocumentationCompletedBy

str?

Which note closed the chain — attribute a long tail without a second query.

Data-quality signals check these before trusting an aggregate

COLUMN	TYPE	NOTES
SeriesLostToNearerAppt	int	>0 means this appointment competed for series and lost them to a nearer one. Normal for same-day repeat sims; a spike means something else.
NotesLostToNearerAppt	int	Same idea for notes.
ImageCount_AnyMachine	int	Volumetric imaging near the appointment on <i>any</i> machine, including imports.

QUESTIONS IT ANSWERS DIRECTLY

How long does a sim take, end to end?

Distribution of `CheckInToDocCompleteMinutes`. Use the median, not the mean — the tail is long and driven by documentation, not scanning.

Are we overrunning booked slots?

Share of rows where `CTCompleteToScheduledEndMinutes < 0`, trended by month. Gate on `CTTimestampEra == 'Reliable'` or accept corrected values.

Is documentation or scanning the bottleneck?

Stack `CheckInToCTCompleteMinutes` against `CTCompleteToDocCompleteMinutes`. Both cross-clock, so trend them post-2026-07-27 only.

How much time goes into writing the notes?

`BothNotes_WriteSecondsTotal`, after dropping rows where any `*_WriteSeconds` is negative or implausibly large.

Do sims get documented at all?

Rate of `HasSimOrSetupNote` by month or by `CompletedByUser`. Read the note-coverage gotcha below before drawing conclusions.

Are patients arriving late, or are we starting late?

`StartDelayMinutes` against `ActualDurationMinutes`. Both ARIA-side, so the full history is usable.

FIVE THINGS THAT WILL BITE YOU

- **A blank Set Up Note has two meanings.** Either it was never written, or it was written outside this visit — the note look-back is one day, so a note drafted days ahead as prep is deliberately excluded. The report cannot tell these apart. Do not read note coverage as a compliance rate.
- **Date of Service is editable; Date Entered is not.** Anything built on `*_DateOfService` can be backdated by the author. The `*_WriteSeconds` columns use it as their start point, which is why negative or hour-scale values appear — those rows were edited, not slowly typed. Screen them.
- **Default date window straddles the clock fix.** Run with no parameters and you get the last 120 days — currently about 2.5 months of pre-fix rows and 1.3 months of clean ones. The pre/post blend shifts every single day, so the same dashboard shows a different mix on each refresh. Pass an explicit start date for anything comparative.
- **NULL is meaningful, not missing.** A NULL `CheckInDateTime` means the patient was never checked in. A NULL note timestamp means no in-scope note matched. Filling these with zeros will silently invent data.
- **Pre-2025-10-01 does not exist here.** The start date is clamped to that floor, because CT image-to-machine attribution effectively began 2025-10-06. Asking for earlier returns data from the floor onward rather than erroring — the window you asked for is not the window you got.

Source: `Production/Simulation_Timing.sql` · one row per completed Lacey CT_Sim

simulation with matched CT imaging

Clock correction is ON — pre-2026-07-27 CT timestamps are algorithmically adjusted;

CTClockOffsetMinutes reports the adjustment per row

Exports contain PHI in PatientId, PatientFullName and AppointmentNote